The Price of Sophistication:
When do spatial and robust models pay in carbon-aware scheduling?

Move work to the cleanest grid and cut 18,000 to 46,000 tonnes CO_2 a year, plus a rule for when extra modelling is wasted

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DATA & TOOLS

Electricity Maps

Real-time grid carbon, 2021 to 2025

Solved with **CVXPY · CLARABEL · HiGHS**

Day-ahead forecast → Migration scheduler → Price each modeling layer → Decision rule

Active inter-region migration cuts tail carbon by 4.0 to 9.9%; the added sophistication (joint covariance, robustness) pays off only at a severity real grids never reach.

THE LEVER	THE SCREEN	THE PRICE	THE LIMIT
4-10%	**~0%**	**M* = 3**	**0 / 17**
Tail carbon cut by sending jobs to the cleanest grid	Extra benefit from correlation and copula models	How extreme an emergency must be for robustness to help	Real grids that ever get that extreme

BACKGROUND

The promise

Modern compute is **flexible**: jobs shift in time and across sites, so deferring work to **cleaner hours and regions** lowers operational carbon. Neighbouring grids move together, so spatial structure looks like free signal a robust scheduler could exploit. **Can measured cross-region correlation actually lower out-of-sample carbon?**

DATA & SETUP

What goes in

Electricity Maps hourly lifecycle carbon, 2021-2025 (**43,824 obs/zone**); train 2021-2024, single **locked 2025 test**. Three grids span the dependence spectrum. **Western US** (CISO/BANC/LDWP/NEVP/AZPS): one shared solar-and-weather system across the Western Interconnection, **strongly correlated**. **Eastern US-Canada** (CA-ON/NYISO/MISO/PJM): an Ontario-anchored belt swept by the same weather fronts, **also strongly correlated**. **Diversified** (CISO/ERCOT/BPAT): solar, wind, and hydro across three interconnections, deliberately misaligned so clean periods never coincide (**near-zero**). California-Nevada and Iberia-France panels corroborate. Facility: R zones x 24 h, 50 MW cap, **80% utilization**, with ramp, deferral, and optional thermal/carbon-budget limits.

METHOD & RIGOR

How it is built and checked

A **Mahalanobis Wasserstein DRO**, a second-order cone program in **CVXPY (CLARABEL / HiGHS)**, minimizing **CVaR** of carbon cost (*Rockafellar-Uryasev 2000*):

$$\min_{x\in\mathcal{X}} \; \langle \bar{\rho}, x\rangle + \varepsilon\sqrt{x^\top \hat{\Sigma} x}$$

$\langle \bar{\rho}, x\rangle$	ε	$\sqrt{x^\top \hat{\Sigma} x}$
expected carbon cost: the dominant mean field	ambiguity radius: how much we robustify	robustness penalty; $\hat{\Sigma}$ is the covariance — correlation enters ONLY here

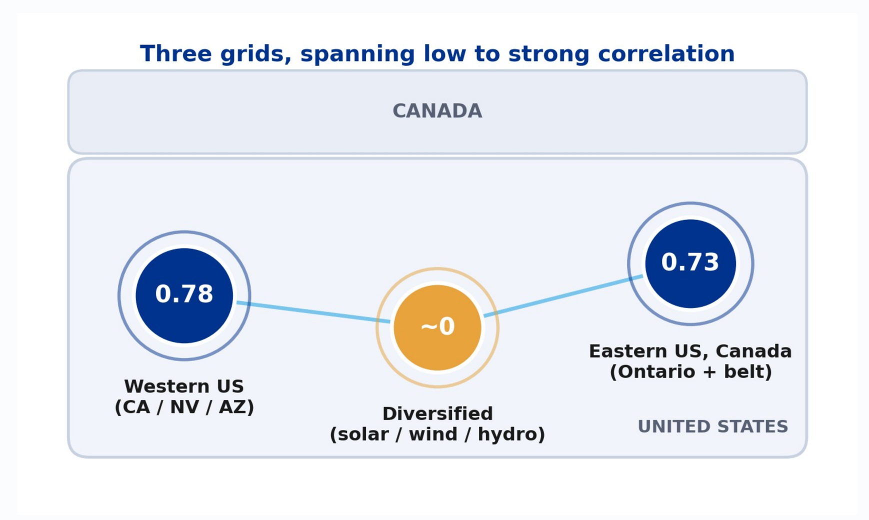
Active transfer adds inter-region flows. Passive cross-region structure enters only through the off-diagonal blocks of Σ, which a **pre-committed shuffled-marginals test** zeroes out. Both arms share one feasible set, so any difference is the dependence model, not the constraints.

202 unit tests	green CI on every push
pre-registered	settings locked before the single 2025 test read
bit-reproducible	pinned lockfile; bootstrap CIs, BH correction, walk-forward; every number traced to an archived table

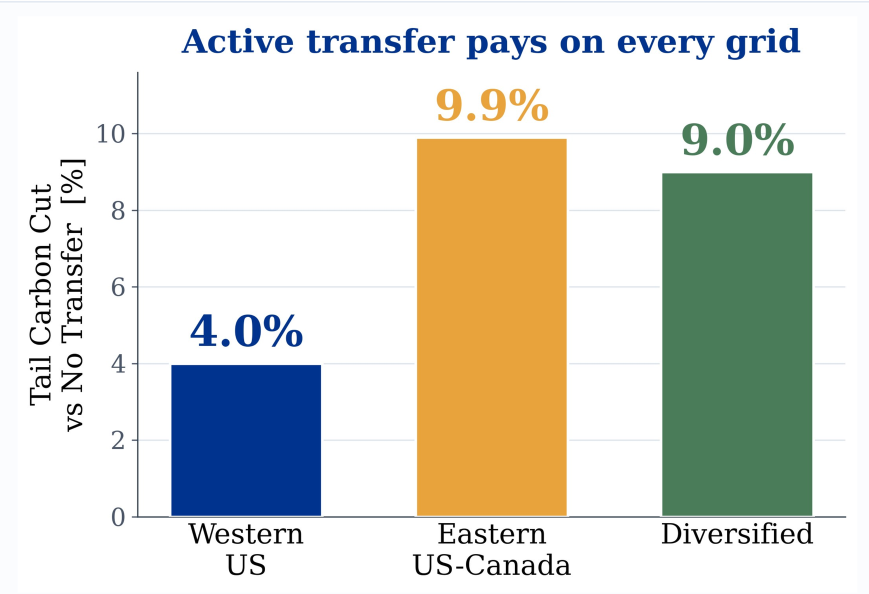
STEP 01 THE LEVER

Moving the work cuts carbon 4 to 10%

A working day-ahead scheduler ships compute to whichever grid is cleanest each hour, scored against an honest no-transfer baseline on the same feasible set. Active migration cuts out-of-sample tail carbon (CVaR, the worst 5% of days) by **4.0 to 9.9%** (Western 4.0, Eastern 9.9, Diversified 9.0) on every grid we tested.



Three grids span the US and Canada, **weak (~0) to strong (0.78)** residual correlation.



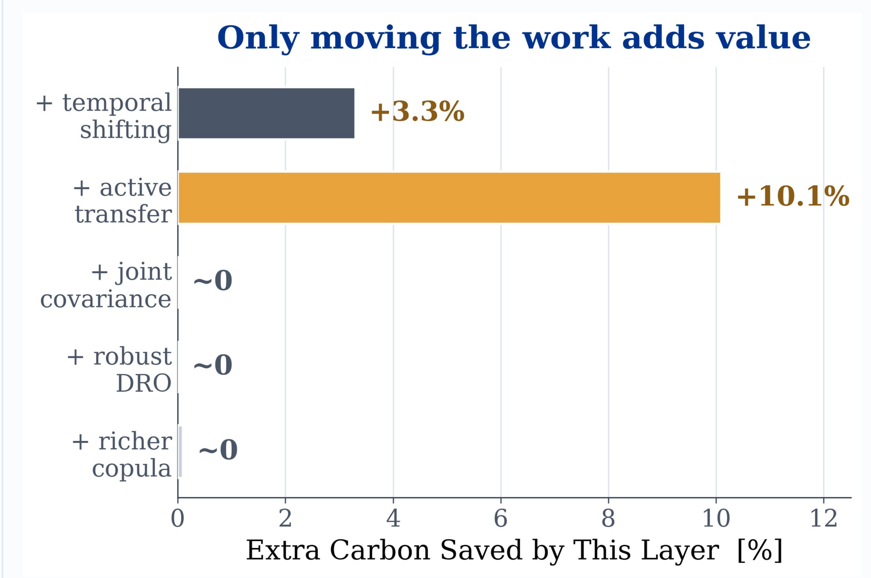
The lever pays on **all three grids**; a small transfer budget already captures most of the cut.

THE VALUE This lever needs no dependence model and no tuning: at every hour one grid is simply cleaner on average, so the scheduler ships work there. The saving rides the predictable daily carbon cycle, not a fragile correlation estimate, so it should carry to other grids and flexible loads. It is the part of the system worth building first.

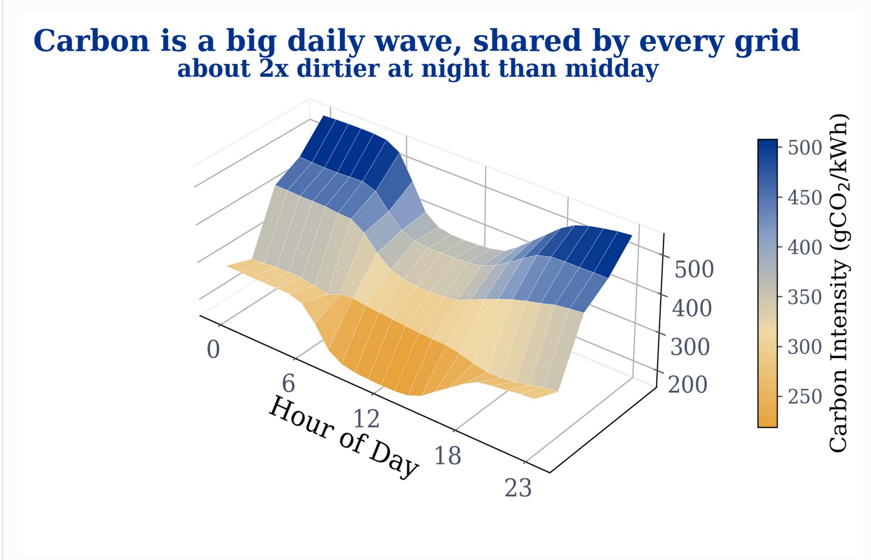
STEP 02 SCREENING RULE

Modelling the correlations adds nothing

Given the lever, does modelling the joint **covariance** or a richer **copula** add value? A pre-committed shuffled-marginals test fits the schedule on the real joint covariance, then again with every cross-region term zeroed, and compares the two out of sample. The answer is no.



Only **transfer** (and temporal shifting) add value; joint covariance, the robust DRO, and copulas each add ≈ 0.



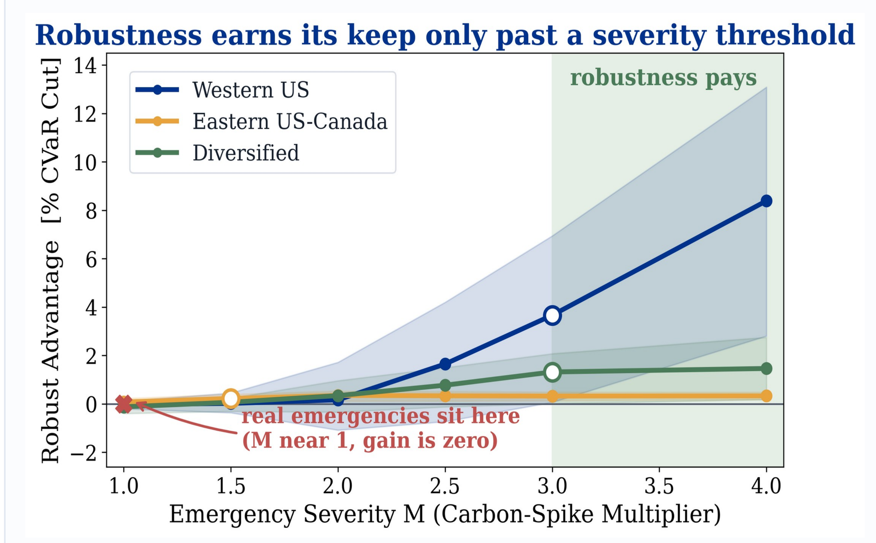
Why: the **mean field dominates**, and the covariance is just a thin ripple on top of that big daily swing.

THE RULE The null holds everywhere we stress it: across every covariance estimator, the full risk tail, a walk-forward year, and even the maximal copula. So the practical call is simple: deploy the plain per-region model, and skip the joint-covariance and copula machinery entirely.

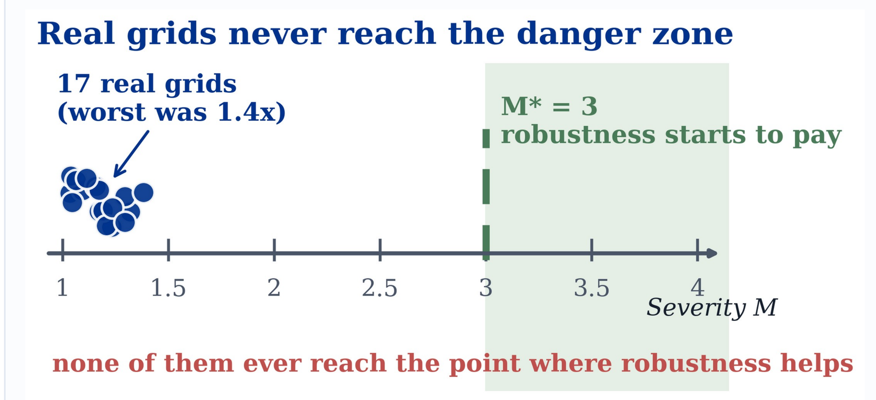
STEP 03 PRICE OF ROBUSTNESS

Robustness only helps in disasters that never happen

Distributional robustness hedges the day-ahead forecast error, trading a slightly higher average for a smaller worst day. We sweep how severe an emergency must be (severity M: 1 is a normal bad day, higher is rarer and harsher) before that trade pays. It only starts to pay above a crossover near **M* = 3**.



Robustness **earns its keep only past M* ≈ 3**; under normal carbon the gain is zero (95% CI bands shown).



Data-grounded emergencies across **17 zones reach only M ≈ 1.4**. Not one of them crosses the threshold.

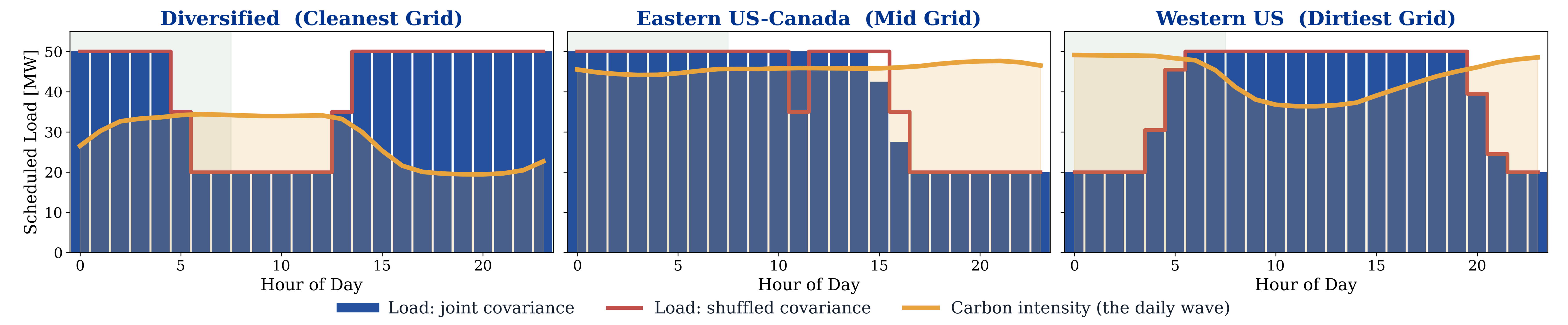
THE LIMIT The protection is real, but it only switches on past a severity the data never reaches. Even Winter Storm Uri peaked near 1.3 times normal, so ship transfer now and turn on robustness only for a shock beyond anything on record.

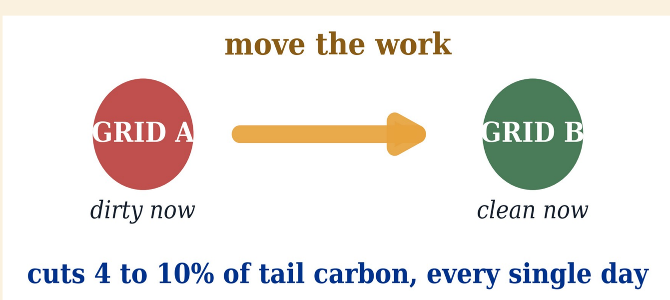
OUR CONTRIBUTION

- a working scheduler with honestly measured savings
- a pre-registered test of validity versus value
- a bound for when richer models cannot help
- a reproducible, pre-registered pipeline (202 tests)

THE RESULT, SEEN DIRECTLY

One zone per grid (cleanest, mid, dirtiest): the joint- and shuffled-covariance plans (navy vs red) overlap almost perfectly. The spatial null, visible by eye.





THE PAYOFF

The value is in physically moving compute toward whichever grid is clean right now. An **active inter-region transfer channel** cuts 4 to 10% of carbon on every grid we tested. For a mid-size facility (about 460,000 tonnes CO_2 a year) that is roughly **18,000 to 46,000 tonnes saved per year**, from code that already runs. The saving comes from the predictable daily carbon cycle, not a fragile correlation estimate, so the same lever should carry to other grids and flexible loads. The heavier statistical layers add nothing until emergencies hit about 3 times normal severity, which real grids never reach.

RECOMMENDATION **Ship the simple migration scheduler now. Skip the joint-covariance and copula modelling, and turn on robustness only if conditions get worse than any real grid has seen. The real contribution is knowing what to leave out, backed by a pre-registered test and a mean-dominance bound.**

REPOSITORY
DATA · THESIS